

Thermal and some physical properties of glasses in the La_2O_3 – CaO – B_2O_3 ternary system

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Abstract

Glasses in the La_2O_3 – CaO – B_2O_3 ternary system were studied. The glass forming range as determined by the appearance of the annealed cast was found to match previously published findings. Clear glasses were formed in the composition range of 5.7–19.1 mol% La_2O_3 with constant B_2O_3 content of 71.4 mol%, and in glasses of constant La_2O_3 : CaO ratio of 1:4 with B_2O_3 content in the range of 71.4–55.0 mol%. The non-linear optical crystalline phase $\text{La}_2\text{Ca}_2\text{B}_{10}\text{O}_{19}$ was crystallized from the clear glasses after heat treatments, as determined by powder XRD. Two types of the LaBO_3 crystalline phases were detected in the partially and the fully crystallized glass compositions outside the glass forming range. Data are reported for the glass transition temperature (T_g), dilatometric softening point (T_d), linear coefficient of expansion (α), onset crystallization temperature (T_x), exothermal peak temperature (T_p), density (ρ) and index of refraction (n_D) in the clear glasses.

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1. Introduction

The superior non-linear optical properties of many borate crystals are of great deal of interest during the last several years in the fields of materials synthesis, development and characterization. These crystals are excellent laser and non-linear optical (NLO) materials that show good second harmonic generation (SHG) efficiency and high damage threshold for laser radiation [1]. Grown single crystals from the ternary systems R_2O_3 – CaO – B_2O_3 (R = rare-earth) of the $\text{RCa}_4\text{O}(\text{BO}_3)_3$ [2,3] and the $\text{R}_2\text{CaB}_{10}\text{O}_{19}$ [4,5] type, are known to be potential candidates for NLO applications. Crystallized glasses containing a crystalline NLO phase can establish multifunctional

glass–ceramic materials of advantageous controlled properties in respect to single crystals [6,7]. Thus, the study of crystallized glasses from the ternary systems R_2O_3 – CaO – B_2O_3 may result with new NLO and laser materials [8,9].

One such possible ternary glass system is La_2O_3 – CaO – B_2O_3 , which was not thoroughly investigated, to the best of our knowledge, regarding the crystalline phases appearing while cooling the melt and after being subjected to heat treatments. The glass forming range for this system exhibits homogeneous glasses in a connecting strip, between the two regions of glass formation in the B_2O_3 – La_2O_3 and B_2O_3 – CaO binary systems (81.0–71.8 and 72.9–58.6 B_2O_3 mol%, respectively) [10]. Red emission under UV source was found in Eu^{3+} doped glass composition of $12.375\text{La}_2\text{O}_3 \cdot 25\text{CaO} \cdot 62.5\text{B}_2\text{O}_3 \cdot 0.125\text{Eu}_2\text{O}_3$, and the fully crystallized product contained a mixed crystalline phases of LaBO_3 along with minor percentage of $\text{CaLaB}_7\text{O}_{13}$ [11]. Subsidiary equilibrium phase relations

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of the $\text{La}_2\text{O}_3\text{--CaO--B}_2\text{O}_3$ system compromise rather complex 18 different three-phase regions [12].

As a potential new NLO and laser materials, we studied the $\text{La}_2\text{O}_3\text{--CaO--B}_2\text{O}_3$ glass system within the compositional range in the vicinity of the glass forming range boundaries. The thermal properties as well as the refraction indices and the density values, were measured for the transparent glasses obtained by conventional melting techniques. The crystalline phases formed during glass making and the crystallized phases appeared after heat treatments of the transparent glasses were detected. The roles of La_2O_3 and CaO in the glass matrix are discussed.

2. Experimental procedures

Analytical grade H_3BO_3 , La_2O_3 and CaCO_3 were used for the synthesis of $\text{La}_2\text{O}_3\text{--CaO--B}_2\text{O}_3$ glasses. The La_2O_3 was heated for 1 h at 1000 °C and was held in a desiccator, to decompose possible hydrates and carbonates, and convert them to the lanthanum oxide. All batches were melted in an electric box furnace at temperatures in the range of 1150–1400 °C in a platinum crucible, until the achievement of homogeneous pourable melts. The melts were cast into pre-heated rectangular box-shaped stainless-steel modular molds to form glass bars. The bars were immediately annealed for 0.5 h in an electric box furnace and then left to cool in the furnace until room temperature was reached. Annealing temperatures (T_{an}) were estimated according to the formula: $T_{\text{an}} = T_g + 15$ °C. Three types of products appeared after casting and annealing the bars: clear glasses, partially-crystallized glasses with visible crystalline phase and fully crystallized glasses. The glass forming range was determined according to the annealed bars appearance. Temperature value accuracy of the electric furnace was within 2 °C range.

Batch of the composition of $13.7\text{La}_2\text{O}_3 \cdot 14.4\text{CaO} \cdot 71.9\text{B}_2\text{O}_3$ was quenched in water at room temperature after melting and homogenizing at 1200 °C.

Powders with average particle size of 200 μm from the annealed clear glasses were measured by means of differential thermal analysis (DTA), at a constant heating rate of 10 °C/min (STD 2960, TA Instruments). From which the glass transition temperature (T_g , i.e., as determined by the inflection point at a temperature difference step change), the onset crystallization temperature (T_x), the glass thermal stability ($\Delta T = T_x - T_g$) and the exothermal peak temperature (T_p) were obtained. The measured values of T_x were used to determine the heat treatments for the crystallization of the clear glasses. The DTA temperatures error ranges were determined from the standard deviations values after running samples of a single glass composition for at least three times, whereas the thermal stability error was calculated using the T_g and T_x average values.

Bulk samples of the annealed clear glasses were measured with an Orton dilatometer (model 1000D) at a heating rate of 3 °C/min. As a result the linear coefficients of thermal expansion in the temperature range of 100–

300 °C (α), the glass transition temperatures (T_g , i.e., as determined by the cross of tangents at slope change) and the dilatometric softening points (T_d) were obtained. Dilatometry measurements accuracy values were taken from the systematic tool sensitivity values.

The density values of the clear glass samples were measured by the Archimedes method using distilled H_2O as the immersion fluid, and the index of refraction values (n_D) were obtained with an ATAGO Abbe refractometer (model 4T). Density error range was determined from the standard deviation of average values from at least three different samples of each glass composition. The n_D values accuracy was taken as the systematic tool sensitivity value.

Powder X-ray diffraction (XRD) measurements were done using a Philips 1050/70 diffractometer ($K_{\text{Cu}\alpha} = 1.540562$ Å) on samples prepared from the crystalline part of the partially and the fully crystallized glasses. In addition, powder XRD measurements were done on samples prepared from the clear glasses after a heat treatment of 1 h at a crystallization temperature above the obtained onset crystallization values (T_x) from the DTA runs. Scanning angles in the 5–70° 2θ range and step size of 0.05° gave the XRD patterns that were used to detect the crystallized phases.

3. Results

3.1. Glass formation range

The glass formation range (GFR) was determined by the appearance of the cast bars from the compositions which are given in Table 1. Fig. 1 shows a ternary diagram of the compositions made from the $\text{La}_2\text{O}_3\text{--CaO--B}_2\text{O}_3$

Table 1
Compositions and annealing temperatures (T_{an}) of glasses investigated in the $\text{La}_2\text{O}_3\text{--CaO--B}_2\text{O}_3$ ternary system

Glass number	Mol%			T_{an} (°C) (for 0.5 h)	Appearance
	La_2O_3	CaO	B_2O_3		
1	13.7	14.4	71.9	675	Clear
2	5.7	22.9	71.4	665	Clear
3	9.5	19.1	71.4	685	Clear
4	14.3	14.3	71.4	685	Clear
5	19.1	9.5	71.4	685	Clear
6	22.9	5.7	71.4	685	Partially crystallized
7	18.6	18.6	62.8	685	Partially crystallized
8	29.7	7.4	62.9	685	Fully crystallized
9	7.4	29.7	62.9	670	Clear
10	9.0	36.0	55.0	670	Clear
11	18.0	27.0	55.0	685	Partially crystallized
12	10.0	40.0	50.0	N/A ^a	Partially crystallized
13	5.0	45.0	50.0	655	Partially crystallized
14	10.0	10.0	80.0	N/A ^a	Two liquids

^a Multiple fractures while casting.

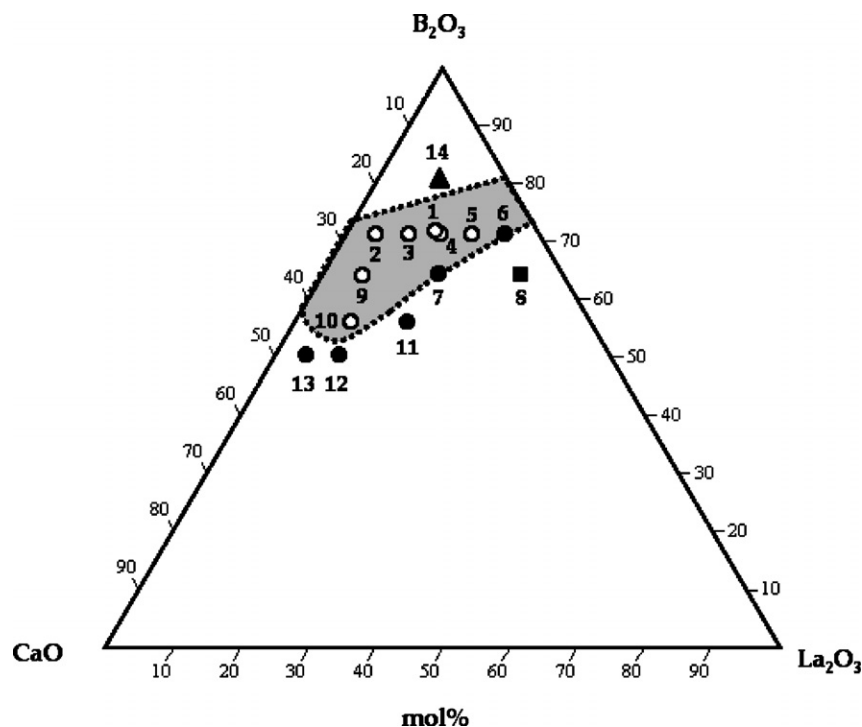


Fig. 1. Glass formation range in the La_2O_3 – CaO – B_2O_3 ternary system as determined by the appearance of the cast bars of various compositions numbered according to Table 1 (filled area – the glass formation range as determined by Imaoka [10]). ○ transparent ● partially crystallized ■ fully crystallized ▲ two liquids.

system. From Fig. 1, compositions of constant 71.4 mol% B_2O_3 form clear glasses with a maximum La_2O_3 content of 19.1 mol% (glass 5). Clear glasses were also obtained from compositions with constant La_2O_3 : CaO ratio of 1:4 and B_2O_3 content in the range of 55.0–71.4 mol% (glasses 2, 9 and 10, Fig. 1).

Partially and fully crystallized glasses were obtained from compositions outside the GFR (as indicated in Fig. 1 and Table 1). Glasses 6, 7, 11 and 13 exhibited both bulk and surface crystallization, whereas glass 12 exhibited surface crystallization only. Glass 8 was fully crystallized. Glass 14 showed a stable phase separation in the form of an outer semi-transparent phase and an inner clear phase. The crystalline phase on the surface of glass 12 and the semi-transparent outer phase of glass 14, both introduced relatively large compression stresses to the clear bulk, therefore, prior the annealing the as cast bars exhibited multiple fractures.

3.2. Differential thermal analysis

All DTA thermograms of the clear glasses exhibited a single exothermal peak, defining the crystallization temperatures range and as shown in Fig. 2. In Table 2 are given the DTA measurements results for the characteristics temperatures values.

Glasses with constant 71.4 mol% B_2O_3 (Fig. 3) showed a decrease of both the T_x values from 821.3 to 783.8 °C and the T_P values from 853.1 to 799.26 °C combined with an

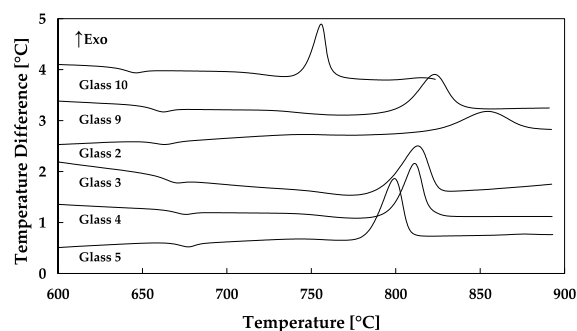


Fig. 2. Powder DTA thermograms of clear glasses 2–5, 9 and 10 (glass numbers according to Table 1) heated at 10 °C/min, from the La_2O_3 – CaO – B_2O_3 ternary system.

increase of the T_g values in the range of 658.0–672.5 °C, as the La_2O_3 content was increased from 5.7 to 19.1 mol%. The yielded ΔT values of these glasses were decreased in the range of 163.3–111.3 °C.

Glass compositions with constant La_2O_3 : CaO ratio of 1:4 and B_2O_3 content in the range of 55.0–71.4 mol% (glasses 2 and 9 and 10, Table 2), showed a decreasing effect in all the temperatures values (i.e., T_g , T_x , ΔT and T_P) with the increasing of La_2O_3 and CaO content (Fig. 4, for the dependence in B_2O_3 content).

3.3. Crystalline phases (XRD)

Based on the onset crystallization (T_x) values of the above DTA measurements (Table 2), bulk samples from

Table 2

Powder DTA results of the clear glasses from the La_2O_3 – CaO – B_2O_3 ternary system, and crystalline phases detected by XRD from samples after 1 h crystallization heat treatment above T_x (in parentheses are errors values, determined as described in the experimental procedure section)

Glass number	T_g (± 0.3) (°C)	T_x (± 0.3) (°C)	T_P (± 0.1) (°C)	ΔT (± 0.4) (°C)	Crystalline phases (after 1 h heat treatment above T_x)	
					Dominant	Minor
1Q ^a	669.5	798.7	813.7	129.2	Not done	
1	668.9	800.7	816.9	131.8	LCB ^b	Undefined residual phase
2	658.0	821.3	853.1	163.3	LCB ^b	– –
3	664.9	792.3	813.1	127.4	LCB ^b	– –
4	670.6	797.1	811.2	126.5	LCB ^b	– –
5	672.5	783.8	799.3	111.3	LaB_3O_6	$\text{Ca}_3\text{La}_2(\text{BO}_3)_4$, LCB ^b
9	657.3	802.8	822.8	145.5	LCB ^b , LaBO_3 ^c	LaBO_3 ^d , Undefined Phase
10	641.0	745.5	755.7	104.5	LCB ^b , LaBO_3 ^c	LaBO_3 ^d

^a As-quenched glass.

^b $\text{La}_2\text{CaB}_{10}\text{O}_{19}$.

^c LaBO_3 monoclinic.

^d LaBO_3 orthorhombic.

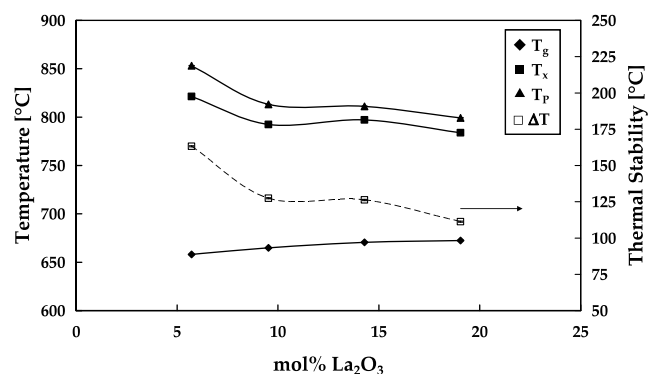


Fig. 3. Effect of La_2O_3 content on T_g , T_x , T_P and $\Delta T (= T_x - T_g)$ in glasses with constant B_2O_3 content of 71.4 mol% measured by DTA. Lines are given as guides to the eyes.

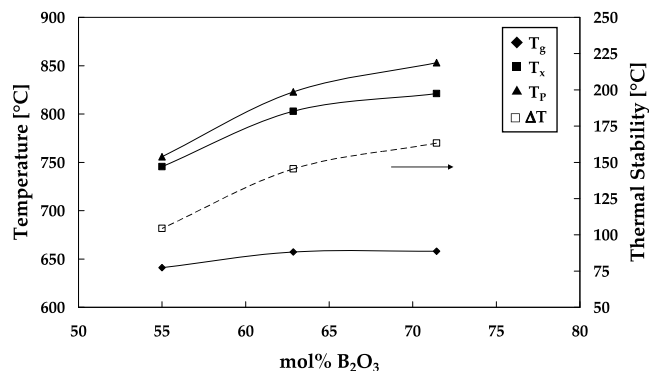


Fig. 4. Effect of B_2O_3 content on T_g , T_x , T_P and $\Delta T (= T_x - T_g)$ in glasses with constant La_2O_3 : CaO ratio of 1:4 measured by DTA. Lines are given as guides to the eyes.

glasses 3–5 and 9 and 10 were crystallized at 815 °C and a sample of glass 2 was crystallized at 850 °C over the period of 1 h. The detected crystalline phases by XRD of the powders from these glasses are also given in Table 2.

XRD patterns of heat treated glasses 1–4 show that the dominant crystalline phase is the monoclinic $\text{La}_2\text{CaB}_{10}\text{O}_{19}$ [4] (LCB) along with some residual undefined phase. Fig. 5 shows the XRD pattern of both the annealed and the heat treated samples of glass 4, indicating LCB crystallization in the heat treated glass. Glass 5 mainly crystallizes to LaB_3O_6 (ICCD 73–1150) with limited counts of $\text{Ca}_3\text{La}_2(\text{BO}_3)_4$ (ICCD 34–1221) and LCB. Glasses 9 and 10 crystallize as a mixture of LCB and LaBO_3 (ICCD 73–1149), both of the monoclinic structure, with minor percentage of LaBO_3 of the orthorhombic type (ICCD 13–0113).

Powder samples from the crystalline segment of glasses with compositions outside the GFR (glasses 6–8 and 11–13, Table 1), were also taken to XRD for the detection of the crystalline phase. Glass 6 showed a mixture of LaBO_3 (orthorhombic) and LaB_3O_6 . It was found that the dominant crystalline phase in glasses 7, 8 and 11 is LaBO_3 (orthorhombic). Glasses 12 and 13 showed crystallization of the monoclinic LaBO_3 . The XRD measurement of powder taken from the semi-transparent outer phase of glass 14 obtained an amorphous pattern, indicating two glassy phases as a result of a stable phase separation.

3.4. Dilatometry

The results of the dilatometry measurements of the clear glass bars are given in Table 3. For glasses with constant B_2O_3 content of 71.4 mol% (glasses 2–5, Table 3), the T_d values were increased in the range of 685–699 °C with increasing La_2O_3 content. These glasses also show a decrease of the α values in the range of 76 – $67 \times 10^{-7} \text{ °C}^{-1}$ (Fig. 6(a)), with increasing La_2O_3 content.

Glass compositions with constant La_2O_3 : CaO ratio of 1:4 and B_2O_3 content in the range of 55.0–71.4 mol% (glasses 2 and 9 and 10, Table 3) show a decrease of the T_d values from 685 to 667 °C and an increase of the α

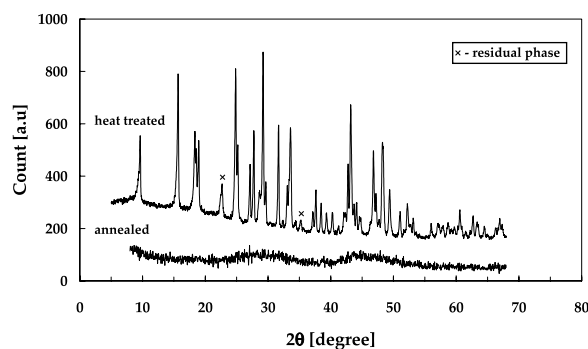


Fig. 5. XRD pattern of the annealed and of the heat treated (1 h at 815 °C) samples taken from glass 4. $\text{La}_2\text{CaB}_{10}\text{O}_{19}$ is the dominant crystalline phase in the heat treated glass.

Table 3

Dilatometry, index of refraction (23 °C) and density results of the clear glasses from the La_2O_3 – CaO – B_2O_3 ternary system (in parentheses are errors values, determined as described in the experimental procedure section)

Glass number	$T_g (\pm 1)$ (°C)	$T_d (\pm 1)$ (°C)	$\alpha \times 10^7 \pm 1$ (°C ⁻¹)	$n_D (\pm 0.001,$ 23 °C)	$\rho (\pm 0.06)$ (g/cm ³)
1	666	697	70	1.682	3.42
2	660	685	76	1.629	3.01
3	668	688	72	1.664	3.04
4	670	696	71	1.692	3.42
5	672	699	67	1.716	3.73
9	653	683	78	1.662	3.34
10	639	667	89	1.687	3.54

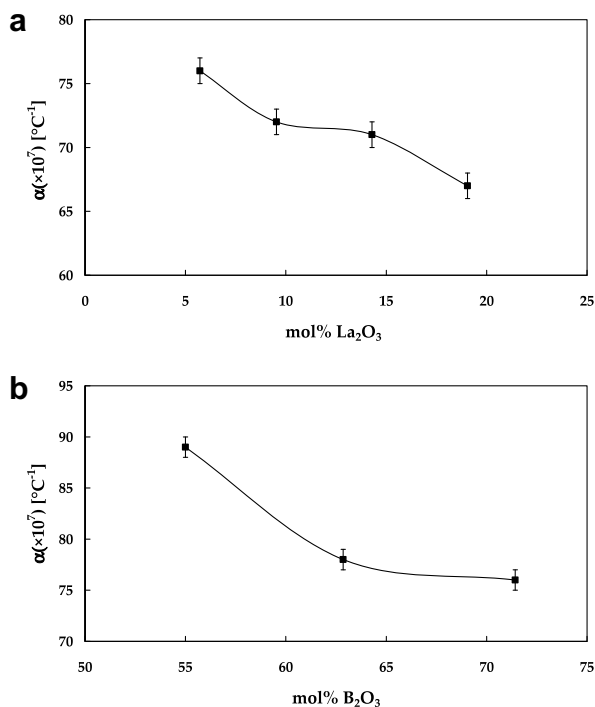


Fig. 6. α Values measured by dilatometry of: (a) glasses with constant B_2O_3 content of 71.4 mol%, (b) glasses with constant La_2O_3 : CaO ratio of 1:4. Lines are given as guides to the eyes.

values from 76 to $89 \times 10^{-7} \text{ °C}^{-1}$ (Fig. 6(b)), as the content of La_2O_3 and CaO was increased.

3.5. Density and refractive index

The results of the index of refraction (n_D) at 23 °C and the density (ρ) measurements of the clear glass bars are given in Table 3. Glass 1 density and n_D values are 3.42 g/cm³ and 1.682, respectively. The density values of glasses with constant 71.4 mol% B_2O_3 (Fig. 5(a)) were increased in the range of 3.01–3.73 g/cm³ as the La_2O_3 content increased from 5.7 to 19.1 mol%. These glasses also show an increase of the n_D values from 1.629 to 1.716, as shown in Fig. 7(a). Glass compositions with constant

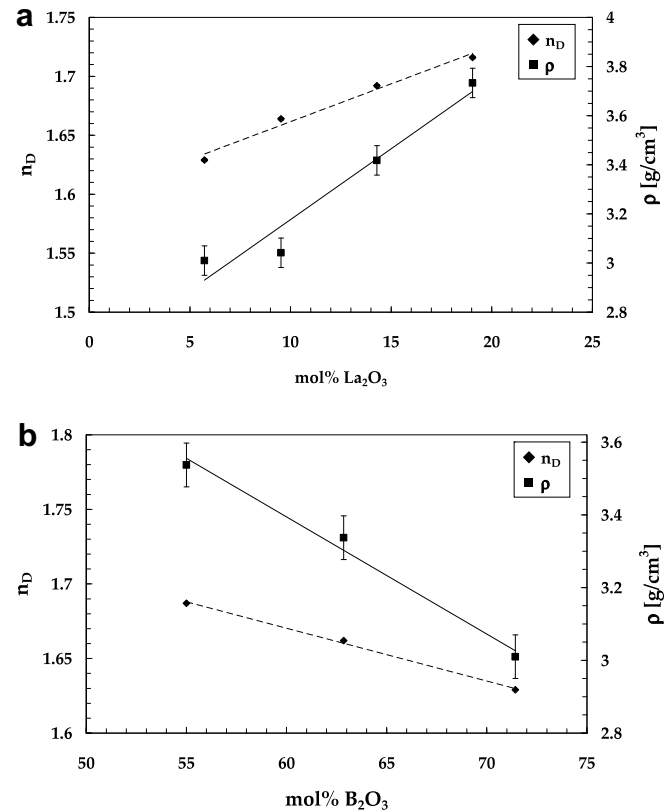


Fig. 7. n_D and ρ values of: (a) glasses with constant B_2O_3 content of 71.4 mol%, (b) glasses with constant La_2O_3 : CaO ratio of 1:4. Lines are given as guides to the eyes.

La_2O_3 : CaO ratio of 1:4 and B_2O_3 content in the range of 55.0–71.4 mol%, show an increase of the density values from 3.01 to 3.54 g/cm³ and an increase of the n_D values from 1.629 to 1.687 with increasing La_2O_3 content (Fig. 7(b)).

4. Discussion

The obtained GFR, as shown in Fig. 1, spreads within the connection strip between the two binary glass sub-systems of B_2O_3 – La_2O_3 and B_2O_3 – CaO , in agreement with the previously determined data of Imaoka [10] and Brow et al. [13].

The formation of the two glassy phases in glass 14 is related to entering the 2-liquids compositional range in the binary subsystems La_2O_3 – B_2O_3 and CaO – B_2O_3 [14], and is supported by the amorphous XRD pattern resulted from the semi-transparent outer phase.

It can be seen from Table 2, that the annealing step effect on the DTA temperatures values of glass 1 (glasses 1 and 1Q, Table 2) is minor. However, in general the T_P and T_x values were shifted to high temperature values in the annealed glass, whereas the T_g values remained unchanged within the experimental error. As a consequence, the yielded thermal stability value ($\Delta T = T_x - T_g$) of 129.2 °C in the as-quenched glass 1 increased to 131.8 °C

in the annealed glass 1. Increasing of the ΔT and the T_p values after glass annealing is an indication for the relaxation of thermal stresses introduced in the bulk during the casting process.

Lowering the glass thermal stability and the exothermal peak temperature, for glasses with constant 71.4 mol% B_2O_3 (Fig. 3), indicates an increased crystallization tendency of the glass matrix as the La_2O_3 content was increased. Consequently, glass composition with 22.9 mol% La_2O_3 (glass 6, Table 1) was partially crystallized during casting and annealing. The observed trends in glass compositions with constant $La_2O_3:CaO$ ratio of 1:4 and B_2O_3 content in the range of 55.0–71.4 mol% (Fig. 4) for the DTA temperatures values, are expected due to the reduced content of the glass forming oxide, B_2O_3 .

From Table 2 it can also be seen that the crystallization of LCB could be achieved from clear glasses of the stoichiometric composition (glass 4, Table 1) as well as with some compositional deviation towards rich CaO content, after a 1 h heat treatment above the onset crystallization temperatures (T_x). The non glass forming compositions (outside the obtained GFR) generally crystallize to $LaBO_3$ while cooling and annealing the cast bars. Moreover, the different crystalline phases in the heat treated glasses (glasses 2–5, Table 2) and in the partially crystallized glass 6, may be correlated to the $BO_4:BO_3$ ratio of borate glasses structural units. Increased La_2O_3 content decreases the relative amount of BO_4 , from the ratio of 3:2 for LCB [4] (glasses 2–4) to 1:2 for LaB_3O_6 (glass 5) [15] and to 0:1 for $LaBO_3$ (partially crystallized glass 6) [16]. The crystallization of LCB only (for glass 2, Table 2) and both $LaBO_3$ and LCB (for glasses 9 and 10, Table 2) with increasing content of La_2O_3 and CaO may indicate that the BO_4 relative amount decreases. Thus, as the La_2O_3 content generally increases the glass matrix might be built from a relatively higher amount of BO_3 (trigonal) structural units than of BO_4 (tetrahedral), implying the formation of a less interconnected network.

Furthermore, considering that LCB ($La_2CaB_{10}O_{19}$) [4] and $CaLaB_7O_{13}$ (ICCD 44–0353, [17]) are both crystalline phases of the same crystal structure and with great similarity of lattice parameters and interplanar spacing, the reliance on XRD measurements solely (as in Section 3.3) might lead to erroneous findings. Thus, an additional detection could be based on obtaining the SHG effect [6] in cases where crystalline LCB exists in the glass. This suggested additional method was successfully tested on glass 1 (Table 2) after several heat treatments [18], as well as the characterization of LCB crystallization kinetics from a glass matrix [19].

The T_g values from the dilatometry measurements of Table 3 are in excellent agreement with T_g values obtained by DTA (Table 2). Discrepancies of lower T_g values in the range of 2–4 °C are related to the different heating rate used in the dilatometer run compared to the DTA (i.e., 3 °C/min as oppose to 10 °C/min) [20]. Thus, the trends of the T_g values in compositions with constant 71.4 mol% B_2O_3 and

with constant $La_2O_3:CaO$ ratio of 1:4, are similar to the obtained ones from the above DTA results.

From the dilatometry results given in Fig. 6 and in Table 3, it can be seen that the observed trends of the α values as well as of T_g and T_d , in the clear glasses with increasing La_2O_3 content, could be explained by the electrostatic characteristics of the systems' cations. Although the ionic radii values of Ca^{2+} is 1.00 Å and of La^{3+} is 1.03 Å (for CN = 6), the ionic field strength with oxygen of La^{3+} is higher than that of Ca^{2+} (0.51 and 0.35, respectively). In addition, both CaO and La_2O_3 demonstrate the same, yet opposite, impact on the α values trends (e.g., CaO tends to increase the α values, where La_2O_3 tends to decrease them) [21]. Therefore, for glasses with constant 71.4 mol% B_2O_3 (glasses 2–5, Table 3), the increase in the La_2O_3 content yields a glass matrix with stronger bonds with the neighboring oxygen ions. Hence, the observed increase of T_g and T_d and the decrease of α in these glasses. For glasses with constant $La_2O_3:CaO$ ratio of 1:4 (glasses 2, 9 and 10, Table 3), the CaO impact on the glass matrix is more pronounced than that of La_2O_3 as indicated by the gradual decrease of B_2O_3 content. Hence, the observed decrease of T_g and T_d and the increase of α in these glasses.

Similar characteristics dependencies were found in other glass systems and compositions with increased La_2O_3 content [22,23], with various $La_2O_3:CaO$ ratios [13] and of increased field strength oxides [24]. Thus, from the data presented in Tables 2 and 3 as well as from Fig. 6, increase of La_2O_3 over CaO served to increase the T_g and the T_d values and to reduce the α values, whereas general increase of CaO displayed an opposite effect, both are characteristics attributed to glass modifying oxides in borate systems [25,26].

In view of the much heavier molar weight of La_2O_3 than either CaO or B_2O_3 , its reflection on the glass density overwhelms any effect of the other oxides of the system [27]. Thus, the density of all the glasses increases with increasing La_2O_3 content (Table 3).

Additionally, structural transformation of borate units from BO_4 to BO_3 (i.e., from the tetrahedral $B\emptyset_4^-$ to the trigonal $B\emptyset_2O^-$, where $\emptyset$ represents bridging oxygens) could thus increase the mean polarizability of the glass, by introducing more polarizable non-bridging oxygens with the increase of the La_2O_3 content [13]. Therefore, glasses with increased La_2O_3 content are expected to have higher mean polarizability values which are reflected in the n_D values, as given in Table 3. This is also in agreement with the empirical correlations of ρ and n_D in glasses [28,29].

5. Summary

The thermal and some physical properties of glasses in the La_2O_3 –CaO– B_2O_3 were investigated. The glass formation range was re-determined and was found to agree with previously published data [10]. The role of La_2O_3 and CaO in the studied glasses structure was attributed to glass modifying oxides. T_g and T_d were found to increase and α , T_x ,

ΔT and T_P were found to decrease with increase of the La_2O_3 content in glasses with constant 71.4 mol% B_2O_3 . A decrease of T_g , T_d , T_x , ΔT and T_P as well as an increase of α were found in glass compositions with La_2O_3 :CaO ratio of 1:4 as the B_2O_3 content was decreased.

LCB crystallizes from near stoichiometric glass compositions (i.e., $14.3\text{La}_2\text{O}_3 \cdot 14.3\text{CaO} \cdot 71.4\text{B}_2\text{O}_3$ mol%) after 1 h heat treatment above the onset crystallization temperatures. LaBO_3 was found to crystallize in glass compositions outside the glass forming range.

It was argued that the BO_3 structural unit amount increases in the crystalline phases in heat treated and partially crystallized glasses, with increasing La_2O_3 content.

Increased La_2O_3 content was found to increase the values of n_D and ρ , in all compositions of the clear glasses.

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